

Instruction manual



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BEHAVIOUR OF COLUMN & STRUTS APPARATUS

INSTRUCTION MANUAL

Behavior of Columns & Struts Apparatus – Working Manual

1.0 APPARATUS REQUIRED:

- Buckling Column Apparatus
- Board : 1 No.
- Graph paper : 4 Nos.
- Stripes : 2 Nos.
- Digital weight sensor

2.0 THEORY:

If compressive load is applied on a column, the member may fail either by crushing or by buckling depending upon its material, cross section and length. If member is considerably long in comparison to its lateral dimensions it will fail by buckling. If a member shows signs of buckling the member leads to failure with small increase in load. The load at which the member just buckles is called buckling or critical load. For a slender column, buckling load is less than the crushing load. The buckling load, as given by Euler, can be found by using following expression:

$$P = \frac{\pi^2 EI}{l_e^2}$$

- where E = Modulus of elasticity
 = $2.0 \times 10^5 \text{ N/mm}^2$ for steel
 I = Least moment of inertia of column section
 l_e = Effective length of column

Depending on support conditions, four cases may arise. The effective length for each of which are given as:

1. Both ends are fixed
 $l_e = L/2$
2. One end is fixed and other is free
 $l_e = L/\sqrt{2}$
3. Both ends are free
 $l_e = L$
4. One end is fixed and other is free

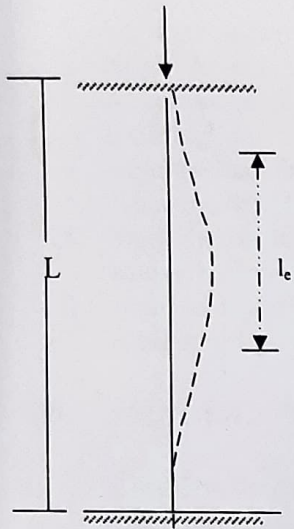
$$l_e = 2L$$

Where, L is distance between points of fixity at top and bottom i.e. unsupported length.

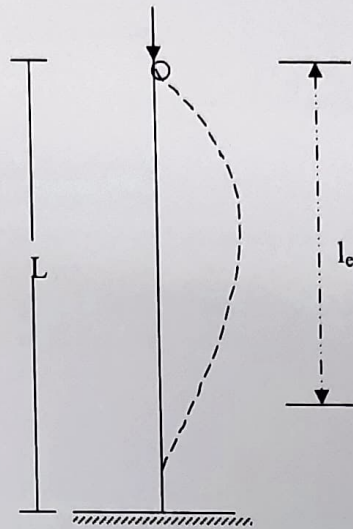
3.0 OBJECTIVE:

To study the behavior of different types of columns and to calculate the Euler's buckling load for each case.

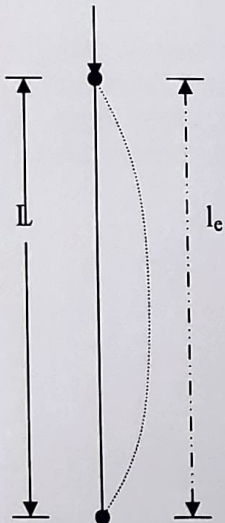
3.0 APPARATUS:



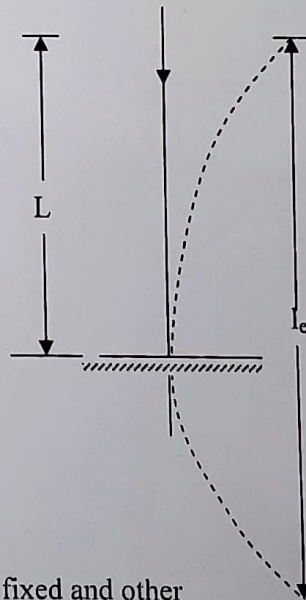
(i) Both ends fixed



(ii) One end fixed other free



(iii) Both end free



(iv) One end fixed and other end free

Apparatus consists of four spring steel columns, which are put along a vertical wooden board. These four columns have different end conditions as below:

1. Both end fixed ($L/2$)
2. One end free and other fixed ($L/\sqrt{2}$)
3. Both ends free (L)
4. One end fixed and other end free ($2L$)

4.0 SUGGESTED EXPERIMENTAL WORK:

1. Pin a graph paper on the wooden board behind the column.
2. Apply the load through wheel increasing gradually.
3. At certain stage of loading the columns shows abnormal deflections and gives the buckling load.
4. Note the buckling load with the help of digital load meter.
5. Trace the deflected shapes of the columns over the paper.
6. Mark the points of change of curvature of the curves and measure the effective or equivalent length for each case separately.
7. Also calculate the theoretical effective lengths and thus buckling loads by the expressions given above and compare them with the observed values.

5.0 RESULTS AND DISCUSSIONS:

1. Calculate the Euler's buckling load for each case.
2. Also calculate the theoretical effective lengths and thus buckling loads by the expressions given above and compare them with the observed values.

6.0 SAMPLE DATA SHEET:

Width of strip (mm)	b=	32 mm
Thickness of strip (mm)	t=	1 mm
Length of strip (mm)	L=	332 mm
Least moment of inertia	I	$\frac{bt^3}{12}$

Sl. No.	End conditions	Euler's Buckling load		Effective Length (mm)	
		Theoretical	Observed	Theoretical	Observed
1.	Both ends fixed				
2.	One end fixed and other pinned				
3.	Both ends Free				
4.	One end fixed and other free with roller.				